

Weiting Liu

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EDUCATION

Northeastern University — Boston, MA | Sep. 2024 – Expect May 2027

Master of Science in Computer Science, GPA: 4.0/4.0

Courses: Software Development, Database Systems, Web Technologies, Machine Learning, Data Structures and Algorithms, Object Oriented Design, Building Scalable Distributed Systems, Mobile Application Development, Advanced Software Development, Computer Systems Engineering, Cloud Computing

Savannah College of Art and Design — Atlanta, GA | Jun. 2023

Bachelor of Fine Arts in Animation

Courses: Computer Graphic, Human Computer Interaction, Tech Animation, Programming, interactive design and game development, Collaborative Projects

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, C#, SQL, Go, R, Ruby, Kotlin, Swift, Scala

Frontend: React, Angular, Vue.js, Redux, HTML5, CSS3, SASS, Bootstrap, Tailwind CSS, Material-UI,

Frameworks & APIs: Spring Boot, Node.js, Express.js, RESTful APIs, FastAPI, Flask, ASP.NET Core, GraphQL, Microservices, WebSocket

Databases & Messaging: PostgreSQL, MySQL, Redis, MongoDB, Kafka, Redis, Elasticsearch, Snowflake, DynamoDB, Cassandra

Cloud & DevOps: AWS (EC2, S3, Lambda, EKS), Docker, Kubernetes, CI/CD, GitHub Actions, Git

Design & Creative Tools: Adobe Creative Cloud, Autodesk Maya, Blender, Figma

Tools & Process: Agile/Scrum, Postman, JUnit, pytest

WORK EXPERIENCE

Huibo Robotics Technology Co., Ltd. | Software Engineer Intern | Jiangsu, China | May 2025 – Sept. 2025

- Designed a real-time robot status monitoring dashboard with **React** and **WebSocket**, visualizing joint angles, torque, and error states across fleets.
- Implemented **RESTful APIs** with **Spring Boot** to expose telemetry and control data, enabling downstream clients to query and visualize robot logs.
- Stored robot telemetry logs and event records in **PostgreSQL** with time-series indexing, reducing historical query latency by 45% across dashboards.
- Automated deployment of monitoring services with **Docker** and **GitHub Actions CI/CD** pipelines, ensuring reproducible builds across environments.

Deluxeton Homes LLC | Interior Designer | GA | Jun. 2023 – Aug. 2024

- Developed **Python** automation scripts for material cost analysis and reporting, reducing documentation effort across interior design projects.
- Created detailed CAD drawings, floor plans, and 3D models using Autodesk tools, translating client requirements into visual design specifications.
- Collaborated with clients and stakeholders to present design concepts, gathering feedback and refining proposals through iterative communication.

Deluxeton Homes LLC | Web Developer | GA | May 2022 – Aug. 2022

- Developed responsive web pages with **HTML**, **CSS**, and **Bootstrap**, implementing color systems and typography from design specs across interfaces.
- Developed reusable page templates and UI components with **JavaScript**, standardizing shared elements to improve consistency across multiple pages.
- Built back-end functionality with **Node.js** and **MongoDB**, implementing APIs and data models to support dynamic web content and product data.
- Coordinated with the product and business team on web asset art direction, ensuring visual brand consistency across all layouts and digital deliverables.

Dezure Development | Architectural Intern | GA | May 2021 – Aug. 2021

- Produced concept sketches and designs from client requirements, translating briefs into visual proposals for architectural review and approval.
- Researched construction materials, dimensions, and specifications, identifying potential structural issues during early design and planning phases.

First Fine Art & Design Academy | Teaching Assistant | GA | May 2020 – Aug. 2020

- Provided 1-on-1 guidance to students on technical animation concepts, identifying individual learning gaps and adapting instructional approaches.
- Assisted professors in developing materials and managing classroom engagement across group and individual art technique learning sessions.

PROJECTS

Secure Reliable File Transfer | Python, Raw Sockets (IP/UDP), AES-256-GCM, HMAC/HKDF-SHA256, Multithreading, Concurrency, AWS

- Engineered a reliable transport protocol from scratch over raw UDP sockets, hand-assembling IP/UDP headers and a custom application header; verified byte-for-byte SHA-256/MD5 integrity across AWS EC2 over ~1.05M packets per 1 GB transfer.
- Implemented Go-Back-N ARQ on a multithreaded send/ACK/retransmit pipeline, sustaining zero-corruption delivery under packet loss.
- Encrypted every datagram PSK handshake, HKDF-SHA256 key derivation, AES-256-GCM authenticated encryption, binding packet metadata as AAD.
- Hardened the protocol with replay protection and hash verification; added AIMD with a dynamic window and a Jacobson/Karn adaptive RTT.
- Implemented Selective Repeat with SACK-style ACKs for targeted retransmission; extended the server to handle concurrent multi-client transfers.

Quora Duplicate Question Detection | Python, PyTorch, scikit-learn, Hugging Face, NLTK, Gensim, Word2Vec/GloVe

- Built and benchmarked multiple duplicate-detection models; achieving 81.67% accuracy and 0.90 AUC-ROC over majority-class.
- Engineered an NLP preprocessing pipeline and 5 handcrafted similarity features fused with ~50K TF-IDF features to drive the top classifier.
- Trained a LSTM (PyTorch, frozen GloVe) reaching 79.2% accuracy and top recall, plus a fine-tuned DistilBERT; benchmarked Word2Vec vs. GloVe.
- interactive CLI across all models, backed by a FastAPI inference endpoint for real-time duplicate-pair testing.

Kambaz Full-Stack Learning Management System | Next.js, React, TypeScript, Redux Toolkit, Node.js, Express, MongoDB Atlas, REST

- Developed a full-stack web application with role-based access control using React/TypeScript for frontend.
- Designed RESTful CRUD APIs with Node.js/Express backed by MongoDB Atlas following clean architecture principles for separation of concerns.
- Managed client state with Redux Toolkit and role-guarded routing; built a responsive React-Bootstrap UI and deployed frontend and backend to production under Git version control.
- Secured the API with JWT authentication and bcrypt password hashing; added Jest + React Testing Library and Supertest suites plus a GitHub Actions CI/CD pipeline.

Memer – Native iOS Meme Generator & Social App | Swift, UIKit, Firebase (Auth / Firestore / Storage / Cloud Messaging), MVC

- Built a full-stack iOS meme-generator and social app in Swift/UIKit (MVC, 15+ screens) on Firebase Auth, Cloud Firestore, and Storage.
- Engineered a gesture-driven editor with draggable, pinch-scalable, flattening layers into a single exported image via UIGraphicsImageRenderer.
- Powered real-time reactions and friend-to-friend chat with Firestore snapshot listeners; added camera / photo-library capture and a SessionManager caching layer for fast profile loads.
- Integrated Sign in with Apple (OAuth) and Firebase Cloud Messaging push notifications; implemented paginated infinite scroll on the trending feed.
- Added XCTest unit and XCUITest UI test suites with a GitHub Actions CI pipeline running on every push.